WORKSHEET

Chapter 6: Polynomials with real coefficients

1 Solve the equation $z^3 + 6z^2 + 6z + 5 = 0$.

2 Solve the equation $3z^3 - 13z^2 + 43z - 13 = 0$.

3 Solve the equation $2z^4 - 7z^3 + 14z^2 - 15z + 6 = 0$.

4 Solve the equation $-3z^4 - 8z^3 - 12z^2 - 12z - 5 = 0$.

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5 Solve the equation $z^5 - z^4 + 7z^3 - 7z^2 + 12z - 12 = 0$.

6 Solve the equation $12z^5 - 20z^4 + 19z^3 - 6z^2 - 2z + 1 = 0$.

7 Express $3z^3 - 13z^2 + 43z - 13$ as a product of linear factors.

8 Express $4z^4 - 4z^3 + 13z^2 - 12z + 3$ as a product of linear factors.

9 Express $z^5 - z^4 + 7z^3 - 7z^2 + 12z - 12$ as a product of linear factors.

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10 Express $z^4 - 2z^3 + 27z^2 - 2z + 26$ as a product of linear factors.

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Answers

1 $z = -5, -\frac{1}{2} \pm \frac{i\sqrt{3}}{2}$

2 $z = \frac{1}{3}, 2 \pm 3i$

3 $z = 1, \frac{3}{4} \pm \frac{i\sqrt{39}}{4}$

4 $z = -1, -\frac{1}{3} \pm \frac{i\sqrt{14}}{3}$

5 $z = 1, \pm 2i, i\sqrt{3}$

6 $z = -\frac{1}{3}, \frac{1 + i\sqrt{3}}{2}, \frac{1 - i\sqrt{3}}{2}$

7 $(3z - 1)(z - 2 - 3i)(z - 2 + 3i)$

8 $4\left(z - \frac{1}{2}\right)\left(z - \frac{1}{2}\right)(z + i\sqrt{3})(z - i\sqrt{3})$

9 $(z - 1)(z + i\sqrt{3})(z - i\sqrt{3})(z - 2i)(z + 2i)$

10 $(z - 1 - 5i)(z - 1 + 5i)(z - i)(z + i)$

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